



# NAWON PERERA

## SOFTWARE ENGINEER

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**GitHub** - <https://github.com/nawonperera>

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### SUMMARY

Software Engineer with 1.6+ years of experience building applications using .NET, React, and Python. Skilled in developing scalable APIs, fullstack web applications, and data driven solutions. Experienced with ASP.NET Core, SQL Server, and modern frontend frameworks, with a strong focus on clean architecture and performance. Quick learner with a practical approach to solving real world problems.

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### TECHNICAL SKILLS

- Programming & Scripting: C#, Python (ML & AI workflows), JavaScript/TypeScript, SQL
- Frameworks & Libraries: ASP.NET Core (Web API, MVC, Clean Architecture), Node.js & Express, React, Angular, Redux, Tailwind CSS
- Databases & Caching: SQL Server, PostgreSQL, MySQL, MongoDB, Redis
- Cloud & DevOps: Docker, Azure Kubernetes Service (AKS), CI/CD Pipelines
- Messaging & Distributed Systems: RabbitMQ, Ocelot API Gateway, Microservices Architecture
- Tools & Practices: Git/GitHub, REST APIs, JWT Authentication, ASP.NET Identity, Swagger, Postman, AutoMapper

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### PROJECTS

- **Cristal Beauty – E-Commerce API & Web App**

<https://github.com/nawonperera/crystal-beauty-clear-frontend>

<https://github.com/nawonperera/crystal-beauty-clear-backend>

RESTful backend for an online beauty store with JWT/Google OAuth authentication, product catalog, stock tracking, orders, customer messaging, wishlist, and OTP email verification. Admins manage users/products and reply to inquiries. Fullstack web app features a customer storefront and admin dashboard for product/order/user management.

**Tech:** C#, ASP.NET Core Web API, MongoDB, React, Vite, Tailwind CSS, Node.js, JWT, Google OAuth, MailKit, Nodemailer, Supabase

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- **CulinaryCore API**

[https://github.com/nawonperera/CulinaryCore\\_API](https://github.com/nawonperera/CulinaryCore_API)

Restaurant backend with layered architecture, menu management with image uploads, order processing with status tracking, and role-based JWT authentication. Secured endpoints for admin/customer operations.

**Tech:** C#, ASP.NET Core 9, EF Core, SQL Server, ASP.NET Identity, JWT

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- **Villa Management System**

[https://github.com/nawonperera/Villa\\_REST-API-Backend](https://github.com/nawonperera/Villa_REST-API-Backend)

Automates villa property administration with a robust ASP.NET Core backend API and interactive MVC frontend. Features secure role based user authentication via JWT, session state management, and Identity integration. Supports scalable property and accommodation management with API versioning, distributed caching, and repository patterns for clean architecture.

**Tech:** C#, ASP.NET Core Web API, ASP.NET Core MVC, Entity Framework Core, SQL Server, AutoMapper, Identity, Swagger

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- **EShop Copilot – AI Driven Distributed E-Commerce App**

<https://github.com/nawonperera/eshop-copilot>

IBuilt a modern distributed e-commerce application orchestrated by .NET Aspire, featuring a Minimal API backend and a Blazor Server interactive frontend. Accelerated development by engineering custom AI Copilot agent rules and specialized "skills" to automatically scaffold full-stack domain features, API endpoints, and dashboard widgets. Enforced strict architectural guidelines, automated service discovery, and OpenTelemetry observability across the ecosystem via agent instructions.

**Tech:** C#, .NET 10, .NET Aspire, ASP.NET Core, Blazor Server, Minimal APIs, OpenTelemetry, AI Agent Customization, Prompt Engineering

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- **Scalable Microservices eCommerce Platform**

<https://github.com/nawonperera/dotnet-ecommerce-microservices.git>

Built a full-stack eCommerce platform using microservices to manage users, products, and orders independently.

Implemented polyglot databases, real time message brokering, and distributed caching, deploying the containerized system to Azure Kubernetes Service (AKS) with automated CI/CD pipelines.

**Tech:** C#, .NET 8, Angular, PostgreSQL, MySQL, MongoDB, Redis, RabbitMQ, Ocelot, Docker, AKS, Azure Pipelines

## WORK EXPERIENCE

### Spherehead Technologies (Pvt) Ltd - Software Engineer (Contract)

June 2026 – August 2026

- Built a static company website using React, focusing on responsive UI and performance.
- Developed a static site using Next.js, implementing modern routing and SEO-friendly structure.
- Contributed to a full-stack MERN project, working across MongoDB, Express, React, and Node.js.

### ASN IT - Trainee .NET Software Engineer

September 2025 – May 2026

- Developed and maintained .NET applications, fixing bugs, writing unit tests, and following best practices while collaborating with senior developers and cross-functional teams.
- Built a real-time kitchen display system using .NET MAUI to show live order updates for kitchen operations.
- Contributed to multiple projects including POSPointe .NET and developed a self-service kiosk application using Flutter with PAX payment system integration.

### CB Electric System – Trainee Software Engineer

October 2024 – September 2025

- Developed and maintained internal business applications using C#, ASP.NET, and SQL Server
- Performed database maintenance and query optimization
- Assisted in PLC related projects, including basic PLC programming and system monitoring

### Heineken Lanka, PVT LTD - Intern Project Manager

September 2023 – March 2024

- Built IoT based bottle counter with ESP32 for remote monitoring.
- Led De-stoner Machine Mounting Project, overseeing design and setup.

## EDUCATION

### BSc(Hons) in Electronics and Engineering Management

#### SLTC Research University, Sri Lanka

- **Brain-Controlled Wheelchair**

[https://github.com/nawonperera/Brain\\_Controlled\\_Wheelchair.git](https://github.com/nawonperera/Brain_Controlled_Wheelchair.git)

BCI smart wheelchair for quadriplegics using SSVEP & ML motor imagery. Real-time EEG signal classification with FFT maps brain responses to forward/left/right/stop via double eye blink detection for independent mobility.

**Tech: Python, ML, Raspberry Pi, STM32, iWorx DAQ, NumPy, SciPy, Serial Communication**

- **IoT-Based Beverage Can Monitoring System**

[https://github.com/nawonperera/IoT\\_Based\\_Beverage\\_Can\\_Monitoring\\_System](https://github.com/nawonperera/IoT_Based_Beverage_Can_Monitoring_System)

Monitors temperature and availability of beverage cans inside a fridge in real time. Uses MQTT to transmit data to a mobile app, enabling remote tracking of inventory and status changes over time.

**Tech: MicroPython, MQTT, NodeMCU ESP32, SolidWorks, KiCAD, IoT MQTT Panel**

- **Bio-Inspired Prosthetic Arm**

<https://github.com/nawonperera/Bio-Inspired-Prosthetic-Arm-Controlled-via-Smartphone>

A 3-DOF bio-inspired prosthetic arm that emulates natural human movements. Controlled in real time via smartphone using MQTT over WiFi, featuring smooth PWM transitions, automated motion sequences, and dynamic coordinate predictions for precise positioning.

**Tech: MicroPython, ESP32, MQTT, Servo Motors, 3D Printing, IoT**

- **Sea Voyage Project**

Developed and marketed an eco-friendly, solar-powered boat. Applied business planning, product strategy, and project management to deliver a sustainable, market-ready solution.

**Tech/Skills: Solar-powered engineering, Product strategy, Project management, Sustainability**

### High School

#### Trinity College Kandy, Sri Lanka

## ADDITIONAL SKILLS

Other Tools & Platforms:

- Visual Studio, PyCharm, Visual Studio Code, Git, GitHub Copilot, ChatGPT, Claude

Languages:

- English (Fluent), Sinhala (Native)

## REFERENCES

- References available upon request.